Vineeth Raghav Gururajan

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PROFESSIONAL SUMMARY

Working student candidate for Hardware Architecture for Automotive Electronics. M.Sc. Electrical Engineering student at the University of Stuttgart with 1 year of industry experience in electronic circuit design, hardware–software interface definition (CAN, I2C, SPI, UART), system integration, component evaluation and prototype validation. Structured, detail-oriented working style with strong analytical and problem-solving skills. Motivated to learn functional safety (ISO 26262) and architecture modeling tools. Available up to 20 hours per week.

SKILLS

Hardware and system-level engineering: electronic hardware, circuit design, system-level engineering, interface definition, hardware–software interfaces, system integration, component evaluation and selection, schematics and datasheets, PCB design (Altium), prototype bring-up, validation

Automotive communication protocols: CAN, I2C, SPI, UART

Modeling and simulation: MATLAB, PSpice circuit simulation, battery cell modeling, statistical parameter sampling

Programming and test: Python, C/C++, Verilog, embedded C, test automation, debugging; microcontrollers (ARM, ESP32, Arduino)

Learning goals (not yet used professionally): functional safety and ISO 26262, requirements traceability, Enterprise Architect, Cameo Systems Modeler, MATLAB/Simulink

EXPERIENCE

Electronic Design Intern – FocalSystech Pvt. Ltd., Chennai, India (07/2024 – 07/2025)

- Defined electrical and hardware–software interfaces between sensors, actuators and microcontrollers (CAN, I2C, SPI, UART) for reliable system integration in IoT devices
- Designed and validated electronic circuits for embedded systems; evaluated and selected components against technical requirements, working from schematics and datasheets
- Created PCB layouts in Altium and supported prototype fabrication, bring-up and validation
- Automated testing and performance analysis in embedded C and Python, reducing manual test effort

EDUCATION

M.Sc. Electrical Engineering – University of Stuttgart (10/2025 – expected 2027)

- Intelligent Power Grids, Power Systems Engineering, Battery Modeling, Power Electronics, Control Systems

B.Tech. Electronics and Computer Engineering – SRM Institute of Science and Technology, Chennai (2021 – 2025), GPA 8.7/10

PROJECTS

Battery Cell Modeling – Team project, University of Stuttgart (2026)

- Analysed cell-to-cell variance from electrical, thermal, aging and manufacturing tolerances by simulation; implemented and debugged statistical parameter sampling in the simulation model

64-bit RoBA Multiplier with MAC Unit – B.Tech. final-year project, Verilog (2025)

- Optimised the adder architecture (Han–Carlson instead of Kogge–Stone) for area, power and speed; synthesised and validated the design

LANGUAGES AND OTHER

English: fluent (written and spoken). **German:** A2, preparing for the telc B1 exam. **Certificates:** PSpice Fundamentals (TVS India); Certificate of Training (Livewire Ltd.). **Volunteering:** STEM workshops for underprivileged students; post-cyclone relief in Chennai.